



Ilex asprella aqueous extracts exert in vivo anti-inflammatory effects by regulating the NF- κ B, JAK2/STAT3, and MAPK signaling pathways

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ABSTRACT

Ethnopharmacological relevance: *Ilex asprella* (Hook. et Arn.) Champ. ex Benth. (IA) is a representative medicinal plant from the South of the Five Ridges of China. Its roots (RIA) and stems (SIA) have been traditionally used for the inflammation-related diseases, such as acute and chronic pharyngitis, cough, and sore throats.

Aim of the study: To evaluate the in vivo anti-inflammatory effects of IA extracts to provide evidence for its traditional use and to enhance the knowledge of the medicinal properties of IA.

Materials and methods: Models of xylene-induced ear edema in mice and carrageenan-induced paw edema in rats were used for the pharmacological evaluations. The mice were randomly divided into eight groups (n = 10 per group): a model group, a positive control group [dexamethasone (Dex), 10 mg/kg, intragastrically (i.g.)], RIA aqueous extract groups with three dosages (30, 15, and 7.5 mg/kg, i.g.), and SIA aqueous extract groups with three dosages (60, 30, and 15 mg/kg, i.g.). The rats were randomly divided into eight groups (n = 6 per group): a model group, a positive control group [acetylsalicylic acid (ASA), 300 mg/kg, i.g.], RIA groups with three dosages (80, 40, and 20 mg/kg, i.g.) and SIA aqueous extract groups with three dosages (160, 80, and 40 mg/kg, i.g.). Histological examinations of the ear and paw tissues were observed by hematoxylin-eosin (HE) staining, and neutrophil elastase levels were assessed in ear tissues by immunohistochemical analysis. Serum levels of tumor necrosis factor- α (TNF- α), interleukin-6 (IL-6), and interleukin-1 β (IL-1 β) were measured by ELISAs, and expression levels of TNF- α , IL-6, and IL-1 β in rat paw tissues were measured by RT-PCR. The signal transduction proteins p65, I κ B α , Janus kinase 2 (JAK2), signal transducer and activator of transcription 3 (STAT3), p38 mitogen-activated protein kinases (p38), extracellular regulated protein kinases (ERK1/2), and c-Jun N-terminal kinase (JNK) in the rat paw tissues were investigated by western blot analyses. RIA and SIA were characterized by HPLC and LC-MS analyses, and the components were confirmed by comparison with isolated compounds.

Results: Intragastric administration with RIA (30 mg/kg) and SIA (60, 30 mg/kg) significantly mitigated ear edema in mice. RIA administration at 80 and 40 mg/kg reduced paw edema in rats 2–3 h after injection. SIA administration with 160 mg/kg inhibited paw edema in rats after the injection of carrageenan for 1–4 h, and SIA administration at 80 mg/kg inhibited paw edema after the injection of carrageenan for 2–4 h. Meanwhile, RIA (80, 40 mg/kg) and SIA (160, 80 mg/kg) reduced inflammatory cell infiltration in the ear and paw tissues and

Abbreviations: ASA, Acetylsalicylic acid; DAB, tetrahydrochloride dehydrate; Dex, dexamethasone; ELISA, enzyme-linked immunosorbent assay; ERK 1/2, extracellular regulated protein kinases; HE, hematoxylin-eosin; IA, *Ilex asprella*; IL-6, interleukin-6; IL-1 β , interleukin-1 β ; JAK2, Janus kinase 2; JNK, c-Jun N-terminal kinase; MAPK, mitogen-activated protein kinase; MPO, myeloperoxidase; NF- κ B, nuclear factor kappa B; NSAID, nonsteroidal anti-inflammatory drug; PBS, phosphate-buffer saline; p38, p38 mitogen-activated protein kinases; RIA, root of *Ilex asprella*; RT-PCR, real-time reverse transcription-polymerase chain reaction; STAT3, signal transducer and activator of transcription 3; SIA, stem of *Ilex asprella*; TNF- α , tumor necrosis factor- α

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infiltration of neutrophil leukocytes in rat paw tissues. RIA (80, 40, and 20 mg/kg) and SIA (160, 80, and 40 mg/kg) notably inhibited the increases of TNF- α , IL-6 and IL-1 β in the serum and mRNA expression in the rat paw tissues. RIA (80, 40 mg/kg) and SIA (160, 80 mg/kg) reduced the p-p65/p-I κ B α , p-JAK2/p-STAT3, and p-p38/p-ERK1/2/p-JNK levels in the pathological tissues of the animals. Phenolic acids and triterpenoids likely contributed to the anti-inflammatory activity.

Conclusions: Both RIA and SIA aqueous extracts showed anti-inflammatory effects in vivo in a dose-independent manner (20–80 and 40–160 mg/kg, respectively). The underlying mechanisms are mediated by inhibiting the pro-inflammatory cytokines TNF- α , IL-6, and IL-1 β via regulation of the NF- κ B, JAK2/STAT3, and MAPK signaling pathways. The present results provided pharmacological evidence that stems are alternative medicinal parts of IA but function at different doses. Additionally, this study supports the use of IA as an anti-inflammatory herbal medicine.

1. Introduction

Inflammation is the first response of the immunological defense system to mechanical injuries and other noxious stimuli and is also involved in the pathogenesis of many diseases, such as neurodegenerative, cardiovascular and other serious diseases (Kuncha et al., 2012). Inhibition of production of pro-inflammatory mediators, including multiple cytokines such as tumor necrosis factor- α (TNF- α), interleukin-6 (IL-6), and interleukin-1 β (IL-1 β), alleviated inflammatory diseases (Zhang et al., 2011; Wynn et al., 2010; Kinne et al., 2000). Although nonsteroidal anti-inflammatory drugs (NSAIDs) are commonly used to treat inflammatory diseases, these drugs also exhibit side effects, such as gastric irritation, ulcers and renal failure, with chronic administration, indicating the need for new anti-inflammatory drugs (Cao et al., 2016). Identification and analysis of potential drugs from traditional medicines with a long history of use has become one strategy of interest.

Ilex asprella (Hook. et Arn.) Champ. ex Benth. (IA) belonging the family Aquifoliaceae is a well-known medicinal plant from the South of the Five Ridges of China (Flora of China Editorial Committee of Chinese Academy of Sciences, 1999). It is as a typical representative of Traditional Chinese Medicine for treating the inflammation-related diseases in local, such as acute and chronic pharyngitis, cough, and sore throats (Dai et al., 2014; Dictionary of Chinese medicine, 1975; Food and drug administration of Guangzhou provincial standard for Chinese medicinal materials, 2004; Editorial board of flora of China, 1979). In addition to its role as one of major components in local drinks, such as Shaxi Herbal Tea and Wong Lo Kat Herbal Tea, IA is extensively used as a herb in market medicines, such as Gangmei-Ganmaoling Granules, Waigan Granules, and Qingmei-Ganmao Granules, to treat the common diseases that inflammation is occupying an significant role such as acute and chronic pharyngitis, cough, and sore throats, and therefore has received increased attention (Du et al., 2017; Dictionary of Chinese medicine, 1975).

Previous phytochemical studies showed that the bioactive compounds of IA include primarily triterpenoids and glycosides, phenolic acids, and steroids as a minor component (Dai et al., 2014; Du et al., 2017). As one of the major ingredients, triterpenoids, including ursane-type and oleanane-type triterpenoids, and glycosides were reported to have in vitro anti-inflammatory, antiviral, and antitumor activities; however, in vivo evaluations have not been performed due to the difficulty in obtaining sufficient amounts of the triterpenoids or glycosides (Dai et al., 2014; Du et al., 2017). Previous pharmacological studies showed that IA aqueous or alcohol extracts demonstrated antiviral and antitumor activities and regulated lipid metabolism in vitro and in vivo (Du et al., 2017); however, the in vivo anti-inflammatory activity and its mechanism of action have not been elucidated.

The extensive use of IA and its slow growth have led to the current shortage of wild resources. In the past, only RIA was formally used in medicines, but gradually, the use of SIA has increased in Guangdong and Hunan. At the same time, most studies have focused on RIA, with very few reports on SIA, which makes clinical application, especially

alternative application, difficult as evidence on the pharmacodynamics of this product is lacking (Du et al., 2017). This lack of data, except for some clinical reports and local usage, has prompted our team to carry out a systematic pharmacological evaluation, analysis of chemical composition and quality evaluation of IA. A systematic analysis of IA, this report describes the anti-inflammatory activity of IA in models of xylene-induced mouse ear swelling and carrageenan-induced rat paw edema in vivo and explores its effects on the NF- κ B, MAPK, and JAK2/STAT3 signaling pathways.

2. Materials and methods

2.1. Preparation of IA aqueous extracts

The roots and stems of *I. asprella* (Hook. et Arn.) Champ. ex Benth. were collected from the Grass Lake Planting Base of Meizhou City in Guangdong Province, China, in September 2016. The plant material was authenticated by one of the authors, Prof. Zhengzhou Han (SANJIU Medical & Pharmaceutical, Co.). The voucher specimen [2016GM901 (SIA), 2016GM902 (RIA)] was deposited at the Modern Research Center for TCM, Beijing University of Chinese Medicine, China. The dried roots and stems of IA (each 1.0 kg) were extracted with boiled aqueous solution (3 $\times$ 10 L, 2 h each time). After removal of the solvent under reduced pressure, the RIA extract (34.4 g) and SIA extract (41.1 g) were obtained for pharmacological study and HPLC analysis.

2.2. Experimental animals

Male ICR mice (20 $\pm$ 2 g) and SD rats (200 $\pm$ 20 g) were purchased from Vital River Laboratory Animal Technology, Co., Ltd. (Beijing, China). All animals were maintained in standard housing cages at a constant temperature (25 $^{\circ}$ C) under a 12 h light-dark cycle with free access to food and water. All experimental procedures were reviewed and approved by the Ethics Committee of Beijing University of Chinese Medicine and were in accordance with the guidelines for the Care and Use of Laboratory Animals published by the U.S. National Institutes of Health (NIH Publications No. 85-23, revised 1996). The experimental protocol of the study was approved by the Committee for Ethics in Animal Experimentation of the BUCM with approval number BUCM-4-2016-111022-4022 in accordance to the guidelines of the

Table 1
Primer sequences used in RT-PCR.

Genes	Primer sequence
TNF- α	Forward 5'-CCCCTTTATTGTCTACTCT-3' Reverse 5'-AAAGCCCATTTGAGTCCTTG-3'
IL-6	Forward 5'-AAATAGTCCTTCTCAACCCAA-3' Reverse 5'-CCGAGTAGATCTCAAAGTGAC-3'
IL-1 β	Forward 5'-GAGCCCATCTCTGTGACTC-3' Reverse 5'-TCAGCTCATATGGGTCCGACA-3'
GAPDH	Forward 5'-GCGACTTCAACAGCAACTCC-3' Reverse 5'-CACCTGTGTGCTGTAGCCGT-3'

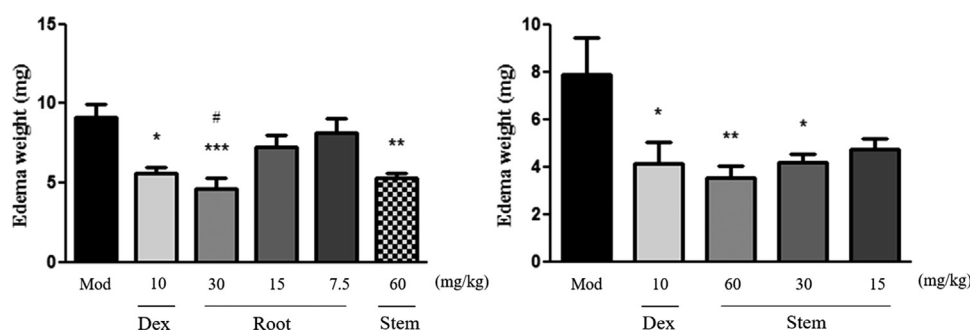


Fig. 1. Treatments with RIA (30, 15, and 7.5 mg/kg) and SIA (60, 30, and 15 mg/kg) significantly reduced the ear edema induced by xylene in mice compared with that of the model group and the positive control Dex group (10 mg/kg). All data are presented as the mean $\pm$ SD per group. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ vs. the model; # $P < 0.05$ vs. the 60 mg/kg stem group.

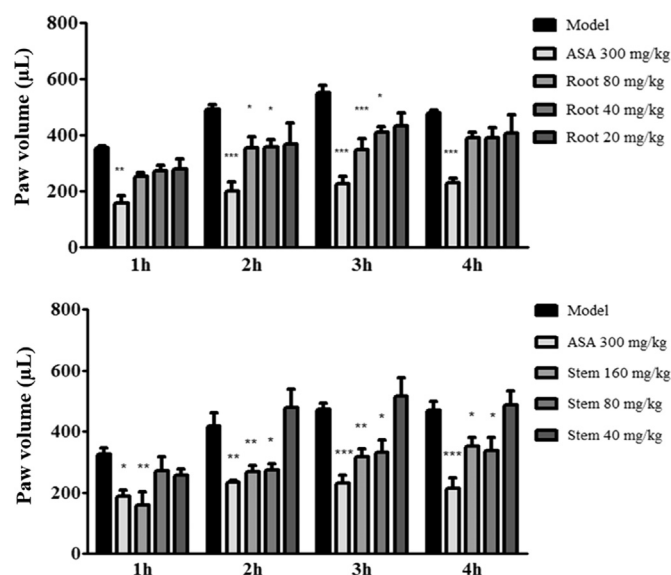


Fig. 2. Treatments with RIA (80, 40, and 20 mg/kg) and SIA (160, 80, and 40 mg/kg) significantly reduced the paw edema induced by carrageenan in a time-dependent manner in rats compared with that of the model group and positive control ASA group (300 mg/kg). All data are presented as the mean $\pm$ SD of six rats per group. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ vs. the model.

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2.3. Xylene-induced mouse ear edema

The xylene-induced ear edema model was generated as previously described (Yang et al., 2017). Mice were randomly divided into a model group, a positive control [dexamethasone (Dex), 10 mg/kg] (Xu et al., 2012; Yang et al., 2017), RIA groups (30, 15, and 7.5 mg/kg), and SIA groups (60, 30, and 15 mg/kg). The model group was intragastrically administered Mill-Q water (0.1 mL/10 g), and 1.5 h later, 30 μ L of xylene was applied to the right ear of each mouse. The left ear served as a self-control. Mice were sacrificed through cervical dislocation, and both ears were removed 30 min after xylene application. Each ear was collected with a cork borer with a diameter of 8 mm and quickly weighed. Edema (%) was calculated using the following formula: edema = right ear weight – left ear weight.

2.4. Carrageenan-induced rat paw edema

The rat hind paw was injected with carrageenan as described previously (Yang et al., 2017). The rats were randomly divided into the control group, model group, ASA as a positive control group (300 mg/kg) (Zhang et al., 2013; Yang et al., 2017), RIA groups (80, 40, and 20 mg/kg), and SIA groups (160, 80, and 40 mg/kg). All groups were treated intragastrically, and the control group received the same

volume of Mill-Q water. A 0.1 mL carrageenan solution (Sigma, 0.9% in control saline) was injected to the plantar side of the right hind paw. Before and after injection of carrageenan for 1 h, 2 h, 3 h, and 4 h, the right paw volumes were measured with a plethysmometer (Anhui Zhenghua Bio Equipment, Co., Ltd., Anhui, China). The inhibition of inflammation was assessed with the increase in paw volumes (paw edema), which was calculated by the following formula: paw edema (%) = $(V_t - V_o)/V_o \times 100\%$, where V_t and V_o represent the final and initial volumes of each rat, respectively.

2.5. Sample collection

Mice were sacrificed through cervical dislocation, and both ears were removed 30 min after xylene application in the model group, positive control group (Dex, 10 mg/kg) (Xu et al., 2012), RIA groups (30, 15, and 7.5 mg/kg), and SIA groups (60, 30, and 15 mg/kg), and right ear tissue slices were immediately fixed in 10% paraformaldehyde solution for 72 h. Then, HE staining and immunohistochemistry were performed.

After carrageenan injection for 4 h, rats were sacrificed, and the right feet of the rats from each group were removed, fixed in 10% paraformaldehyde solution for 72 h, sectioned and stained with HE or frozen with liquid nitrogen and stored at -80°C prior to RT-PCR analysis and protein extraction.

Blood samples were collected from the eye canthus of each group of rats after 4 h of carrageenan injection. The serum was obtained through centrifugation at 3000 rpm for 10 min and stored at -20°C until analysis.

2.6. Histological examination

In the xylene-induced ear-swelling test, biopsies of right ears from all groups were collected, and the tissues slices were fixed in 10% paraformaldehyde solution for 72 h and then stained with HE. In the carrageenan-induced paw edema test, right paw tissues obtained from all the groups were fixed in 10% paraformaldehyde for 72 h, embedded in paraffin, sectioned, and then stained with HE. A light microscope (Leica ICC50HD, Germany) at 40 and 10 $\times$ magnification was used to examine each slide.

2.7. Immunohistochemistry

For immunostaining of the paraffin sections, ear tissues from the xylene-induced ear-swelling test sections were deparaffinized with xylene and rehydrated with ethanol and water. Endogenous peroxidase activity was blocked with 3% H_2O_2 for 15 min at room temperature. Antigen retrieval was accomplished in a microwave oven at 600 W for 15 min in 0.01 M citrate buffer at pH 6.0, and the samples were later rinsed with phosphate-buffered saline (PBS). Sections were incubated with primary anti-MPO antibody at 4°C overnight. The sections were incubated with biotinylated HRP-conjugated goat anti-rabbit antibodies. The colored product was developed by incubation with 3, 3'-

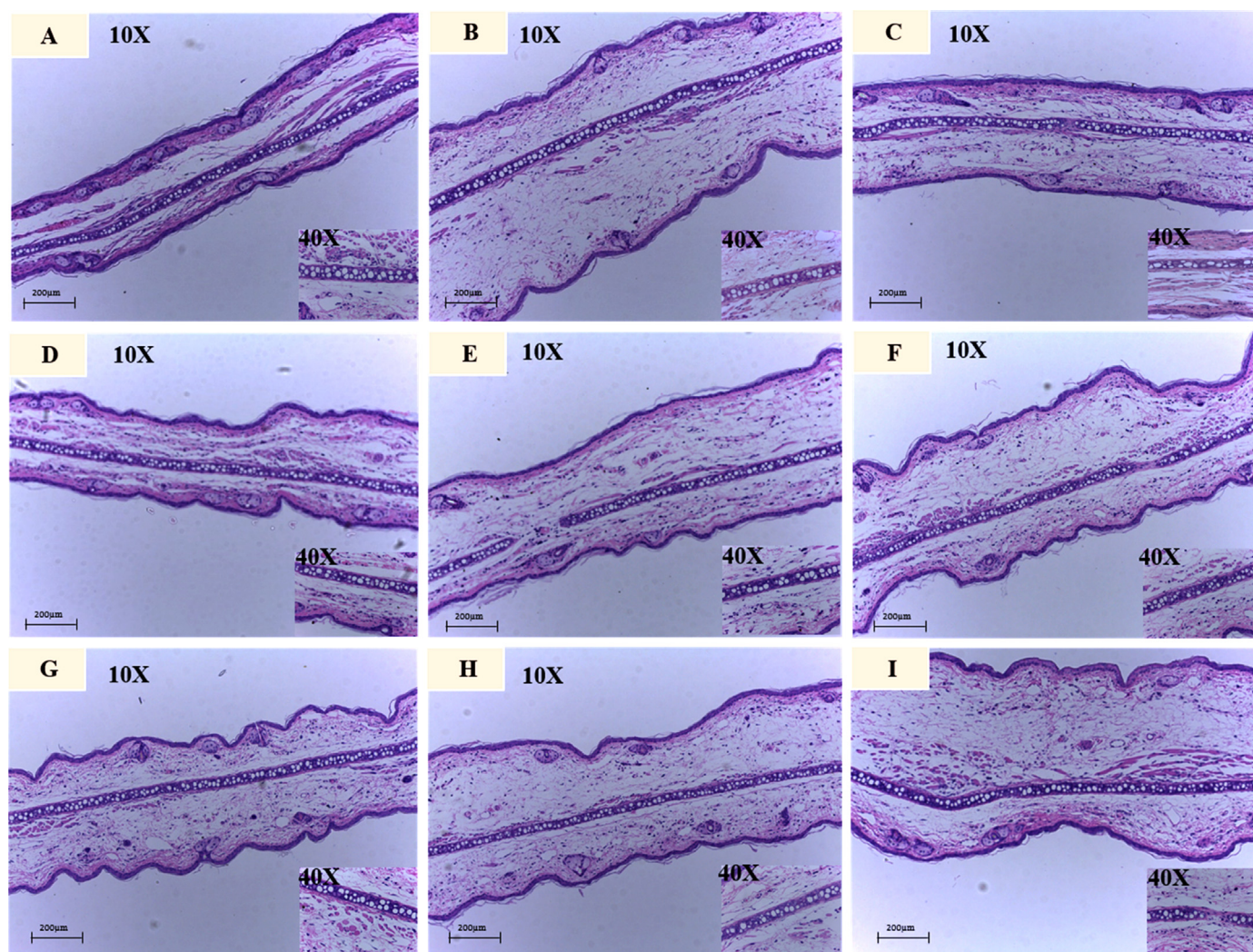


Fig. 3. Histopathological examination of sections of ear tissue treated with xylene (A–I). The neutrophil leukocyte infiltration was reduced and edema was decreased in D–I, and the ear tissues were similar to the control tissues in C.

diaminobenzidine tetrahydrochloride dihydrate (DAB). Immunohistochemical staining was photographed with a light microscope. The intensity of the staining signal was measured and documented using Image-Pro Plus version 6.0 image analysis software.

2.8. Biochemical analysis

Serum levels of TNF- α , IL-6, and IL-1 β were measured by using commercial ELISA kits (Technology Co., LTD., Boster Biological, China), and the results were obtained with an ELISA reader (Perkin Elmer, Enspire). All assays were performed according to the manufacturer's instructions.

2.9. RNA extracts and real-time PCR

Total RNA of the rat paw tissues was extracted and reverse-transcribed into cDNA for 30 min at 42 °C using an RT kit, and the cDNA was amplified using a PCR Supermix kit (TransGen Biotech, Co., Beijing, China). Sequences of the TNF- α , IL-6, IL-1 β and GAPDH primers are shown in Table 1 (Zhu et al., 2016). Amplification reactions were performed for 1 cycle at 95 °C for 1 min, 40 cycles at 95 °C for 5 s and 60 °C for 40 s, followed by 1 cycle of melting curve determination. The amount of mRNA in each sample was calculated and normalized to the level of GAPDH mRNA.

2.10. Western blot analysis

The rat paw tissues were minced, and both RIPA lysis buffer (Applygen Technologies Inc., Beijing, China) and protein phosphatase inhibitor complex (100:1) at 100 mg/mL were added after the samples were disrupted by shaking in a Tissue Lyser II (40,714 Hilden, Qiagen, Germany) at a frequency of 30 times/s for 3 min for homogenization. Then, homogenates were centrifuged at 12,000 rpm for 10 min at 4 °C to obtain the supernatant. Protein quantification was performed with a protein assay kit (Applygen Technologies, Inc., Beijing, China) following the manufacturer's instructions, and then, the samples were boiled for 5 min. The protein lysates (50 μ g) were electrophoresed and separated with 10% SDS-PAGE at 100 V for 2 h and then transferred onto a polyvinylidene difluoride membrane at 300 mA for 1.5. Each membrane was blocked with 5% non-fat milk in TBST and then incubated overnight at 4 °C with the primary antibodies phospho-NF- κ Bp65 and NF- κ Bp65 rabbit mAb (1:1000, Abcam, USA), phospho-I κ B α and I κ B α rabbit mAb (1:1000, Abcam, USA), phospho-JAK2 mouse mAb and JAK2 rabbit mAb (1:1000, Abcam, USA), phospho-STAT3 mouse mAb and STAT3 rabbit mAb (1:1000, Abcam, USA), phospho-JNK and JNK rabbit mAb (1:1000, Abcam, USA), phospho-ERK1/2 and ERK1/2 rabbit mAb (1:1000, Abcam, USA), phospho-38 and p38 rabbit mAb (1:1000, Abcam, USA) and mouse anti-actin mAb (1:5000, Abcam, USA) in 5% non-fat milk with gentle shaking. The following day,

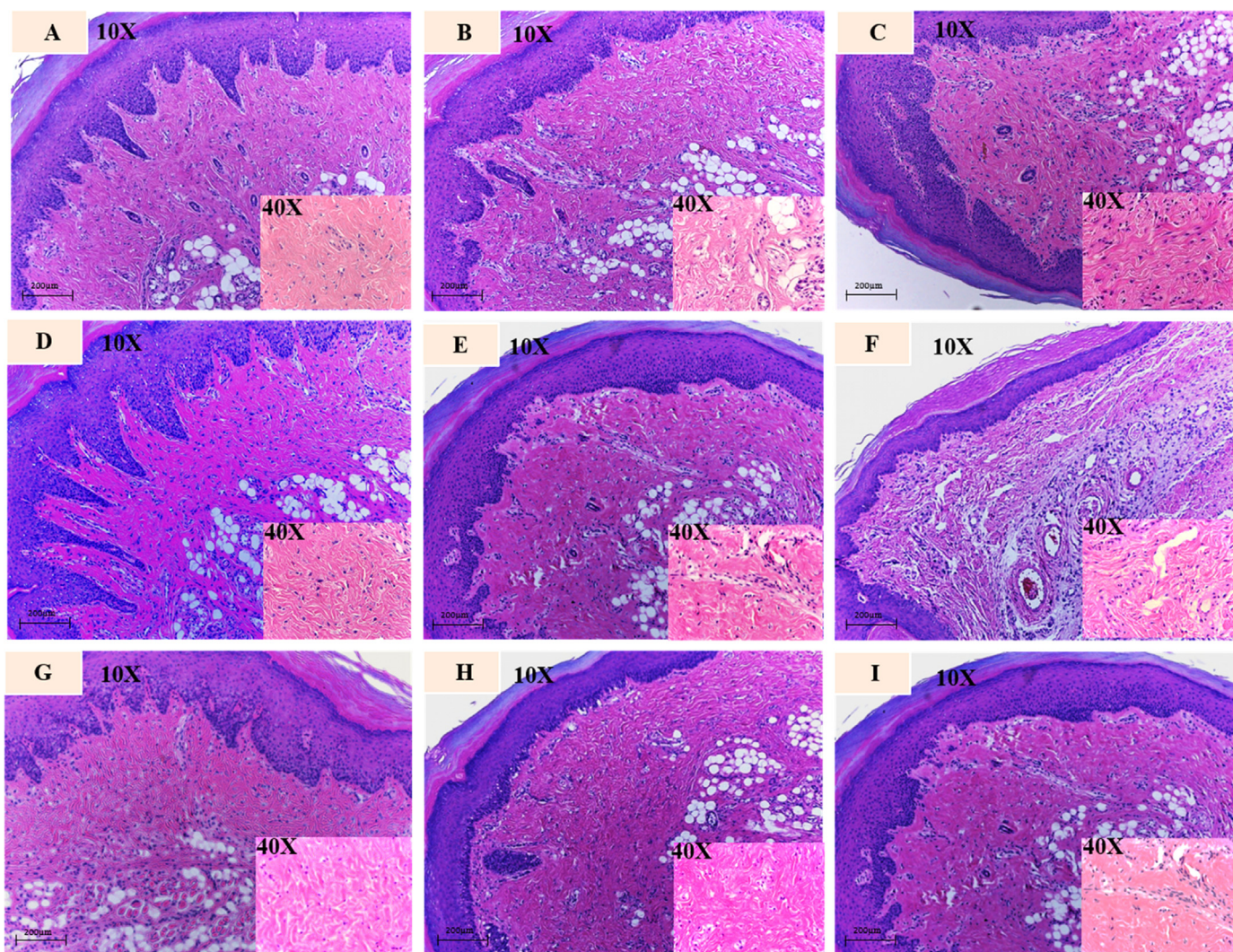


Fig. 4. Histopathological examination of paw tissue sections induced by carrageenan (A–I). The neutrophil leukocyte infiltration was reduced and edema was decreased in D–I, and the paw tissues were similar to the control tissues shown in C.

membranes were washed with TBST thrice for 10 min and then incubated at room temperature for 1 h with the corresponding HRP-conjugated secondary antibody in 5% non-fat milk (secondary antibodies to rabbit monoclonal antibody to p-p65 and p-I κ B α ; p-JAK2, p-STAT3 and p-JNK; p-ERK1/2 and p-p38, 1:2000, Abcam, USA; actin mouse monoclonal IgG, 1:5000, Abcam, USA). Then, the membranes were again washed with TBST and treated with a super ECL kit for visualization. Protein bands were detected using Image Lab Software and quantified with Quantity One 4.6.2 software (Bio-Rad) and one-way analysis of variance (ANOVA) to determine statistical significance.

2.11. Statistical analysis

All data are expressed as the mean $\pm$ SD of at least three independent experiments. Statistical analysis was performed with GraphPad Prism 5.0 and SPSS 20.0 (Chicago, IL, USA) software. One-way ANOVA followed by Dunnett's test was used to evaluate differences among the groups, and differences between two groups were compared by Student's *t*-test. *P* < 0.05 indicated that the difference was significant.

2.12. Instrument, reagents, and HPLC-DAD and HPLC-ESI-IT-TOF-MS analysis

The liquid chromatographic analysis performed on a Shimadzu LC system (Shimadzu, Kyoto, Japan) and the related reagents, including LC-MS grade acetonitrile, ultrapure water, and other analytical-grade solvents, are the same as previously described (Cao et al., 2016; Bai et al., 2017).

The chromatographic separation was performed on an Acquity UPLC BEH C18 column (100 mm $\times$ 3.0 mm, 1.7 μ m, Waters). A 0.1% acetonitrile solution with formic acid (A) and 0.1% aqueous solution with formic acid containing 10 mM ammonium formate (B) were used as the mobile phase, and the flow rate was set at 0.3 mL/min with a gradient program as follows: 0–20 min, 5–25% A; 20–37.5 min, 25–52% A; 37.5–50 min, 52–65% A; 50–55 min, 65–95% A; 55–60 min, 95% A. A 100% portion of the effluent was introduced into the ESI source. UV absorption over 190–400 nm was recorded by the DAD unit, and the detection wavelength was set at 205 nm. The injection volume was set at 2 μ L. At the end of each run, the initial mobile phase solution (5% A) was permitted to re-equilibrate the whole system for 10 min. The optimized operating conditions of high resolution mass spectrometry were as follows: positive mode; nebulizer gas (N₂) flow, 1.5 L/min; CDL temperature, 200 $^{\circ}$ C; heat block temperature, 200 $^{\circ}$ C; detector voltage, 1.40 kV; interface voltage (+), 4.5 kV; ion accumulated time, 10 ms;

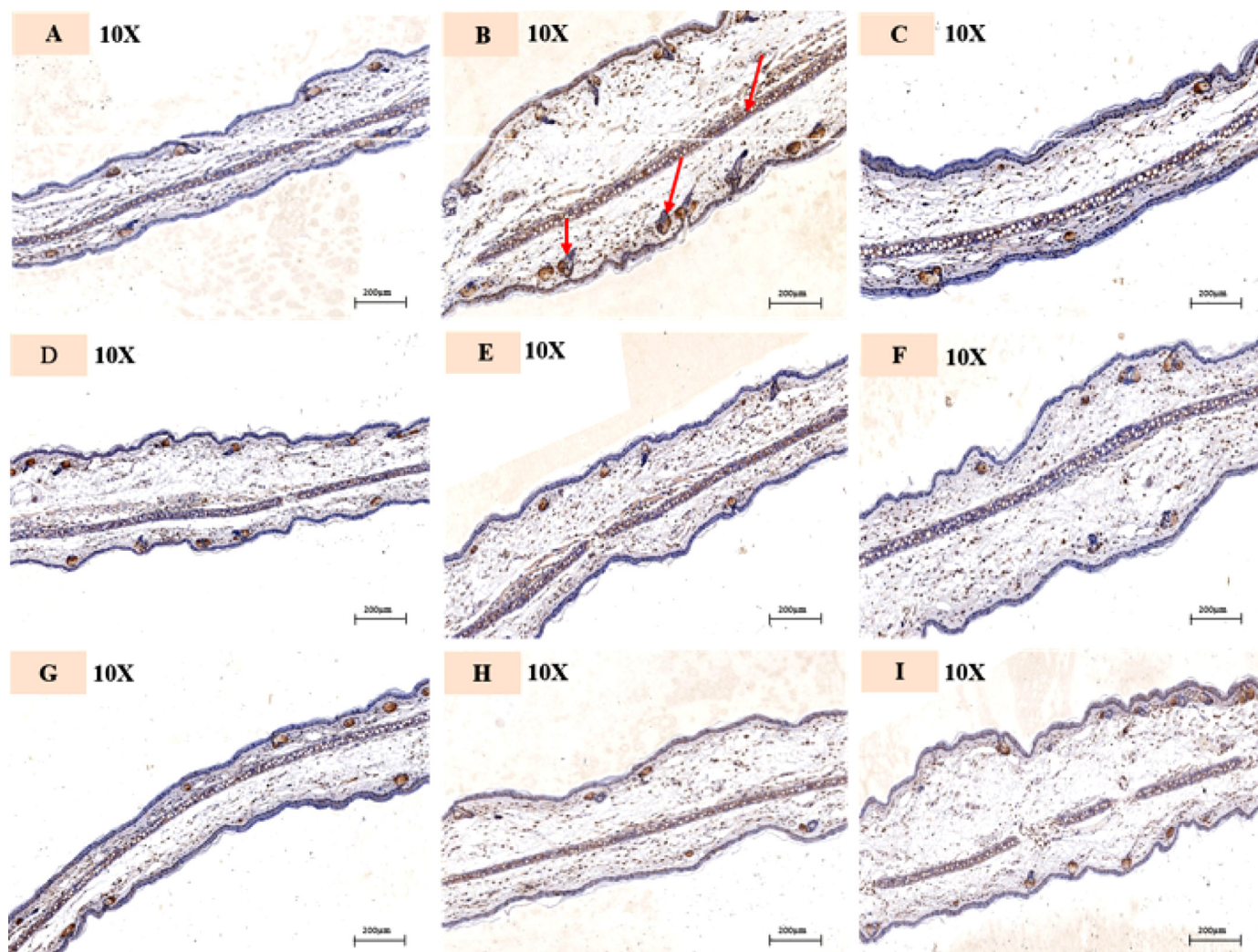


Fig. 5. Immunohistochemical analyses of myeloperoxidase (MPO) in sections of ear tissue induced by xylene (A–I). The red arrow indicates a positive signal. The intensity of the staining signal was measured in each image.

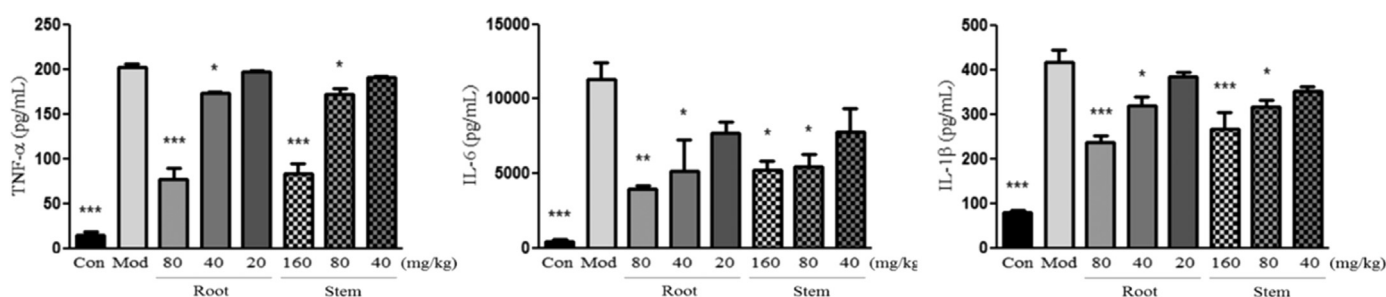


Fig. 6. Results of diagnostic analyses showing serum levels of TNF- α , IL-6, and IL-1 β in different treatment groups. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ vs. the model.

repeat times, 2; collision energy was set at 50% for MS², MS³ and MS⁴; pressure of ion trap, 1.9×10^{-2} Pa; pressure of TOF region, 1.2×10^{-4} Pa; scan range m/z 100–1000; precursor ion isolation, 3.0000 Da. An automatic scan mode was used in the HPLC-ESI-MSⁿ analysis.

The characterization of peaks 1–12 was achieved by analysis of LC-MS data and confirmed by comparison of constituents that were isolated and structurally identified from IA stems with control compounds (Du et al., 2017; Zhang et al., 2018).

3. Results

3.1. Effects of RIA and SIA in the xylene-induced ear edema tests

The RIA and SIA treatments significantly reduced xylene-induced ear edema in a dose-dependent manner, as shown in Fig. 1. RIA at 30 mg/kg and SIA at 60 mg/kg and 30 mg/kg had significant inhibitory effects compared with the treatment in the model group. Interestingly, RIA at 30 mg/kg increased the inhibition rate to 50%, demonstrating a better inhibitory effect than SIA at 60 mg/kg. These findings indicated that RIA had a stronger anti-inflammatory effect than SIA in the ear

Table 2
TNF- α /IL-6/IL-1 β quantification by RT-PCR in paw tissues.

Group	Final concentration (mg/kg)	Relative TNF- α mRNA expression	Relative IL-6 mRNA expression	Relative IL-1 β mRNA expression
Control	–	0.020 $\pm$ 0.003 ^{***}	0.043 $\pm$ 0.001 ^{***}	0.002 $\pm$ 0.000 ^{***}
Model	–	100.000 $\pm$ 0.000	100 $\pm$ 0.001	100 $\pm$ 0.000
Stem	160	0.069 $\pm$ 0.025 ^{***}	0.380 $\pm$ 0.150 ^{***}	0.008 $\pm$ 0.002 ^{***}
	80	12.136 $\pm$ 4.415 ^{***}	0.908 $\pm$ 0.013 ^{***}	0.027 $\pm$ 0.007 ^{***}
	40	17.619 $\pm$ 0.305	26.863 $\pm$ 3.702 ^{***}	0.055 $\pm$ 0.021 ^{***}
Root	80	0.080 $\pm$ 0.001 ^{***}	0.961 $\pm$ 0.949 ^{***}	0.002 $\pm$ 0.001 ^{***}
	40	41.846 $\pm$ 6.606 ^{***}	9.099 $\pm$ 2.284 ^{***}	0.042 $\pm$ 0.005 ^{***}
	20	85.148 $\pm$ 11.421	99.840 $\pm$ 0.840	33.840 $\pm$ 3.272

Values are expressed as the mean $\pm$ SD. Data were obtained from three independent experiments.

** $P < 0.01$.

*** $P < 0.001$ vs. the model group.

edema test.

3.2. Effects of RIA and SIA in the carrageenan-induced paw edema test

As shown in Fig. 2, RIA and SIA significantly decreased paw edema induced by carrageenan in a time-dependent manner compared with that of the model group. The results showed that RIA at 80 and 40 mg/kg reduced carrageenan-induced paw edema after injection for 2–3 h ($P < 0.05$, $P < 0.001$). SIA at 160 mg/kg had a significant inhibitory effect after injection for 1–4 h ($P < 0.05$, $P < 0.01$); the 80 mg/kg dose also inhibited paw edema after the injection of carrageenan for 2–4 h ($P < 0.05$).

3.3. Effects of IA on ear tissue pathology

Inflammatory cell infiltration and edema resulted an increase in ear slice thickness, as shown by the HE method (Fig. 3). HE staining showed less neutrophil leukocyte infiltration and well-structured tissue architecture in the control group (Fig. 3A) compared to the model group, which had extensive dense neutrophil leukocyte infiltration and increased ear thickness (Fig. 3B); the Dex group (Xu et al., 2012) (Fig. 3C) and the RIA (30, 15, 7.5 mg/kg) (Fig. 3D–F) and SIA (60, 30, 15 mg/kg) (Fig. 3G–I) groups showed protection against edema in the ear tissue.

3.4. Effects of IA on paw tissues pathology

As shown in Fig. 4A, no neutrophil leukocyte infiltration and well-structured tissue architecture were observed in the control group histopathology analysis. After carrageenan injection, dense neutrophil leukocyte infiltration, bleeding, muscle fiber breakage, and complete destruction of the foot tissue architecture were observed (Fig. 4B). Treatment with ASA (Zhang et al., 2013) decreased neutrophil leukocyte infiltration compared to that of the model group (Fig. 4C). Treatments with RIA (80, 40, 20 mg/kg) (Fig. 4D–F) and SIA (160, 80, 40 mg/kg) (Fig. 4G–I) alleviated the pathological changes, resulting in little neutrophil leukocyte infiltration and improved structure.

3.5. Immunohistochemical analyses of myeloperoxidase after IA treatment in the model of ear edema

To determine the neutrophil levels, we performed immunohistochemical analysis to assess MPO expression (Zhang et al., 2017). The control group treated with PBS showed low levels of neutrophils (Fig. 5A). Ear tissues demonstrated infiltration by neutrophils in the model group (Fig. 5B). Dex (10 mg/kg, Xu et al., 2012) (Fig. 5C), RIA (30, 15, 7.5 mg/kg) (Fig. 5D–F) and SIA (60, 30, 15 mg/kg) (Fig. 5G–I) alleviated the pathological changes, resulting in less neutrophil leukocyte infiltration than that in the model group.

3.6. Effects of IA on TNF- α , IL-6, and IL-1 β levels in SD rat serum

Serum levels of TNF- α , IL-6, and IL-1 β in the model group were significantly higher than those in the control group ($P < 0.001$) and were decreased in the RIA (80, 40 mg/kg) and SIA (160, 80 mg/kg) groups (Fig. 6).

3.7. Effects of IA on mRNA expression levels of TNF- α , IL-6, and IL-1 β in rat paw edema

Inflammatory cytokine mRNA levels of TNF- α , IL-6, and IL-1 β in the model group were significantly increased compared to those in the control group ($P < 0.001$). A significant decrease in these levels was observed in paw tissue homogenates treated with RIA (80, 40, 20 mg/kg) and SIA (160, 80, 40 mg/kg), and these tissues had less neutrophil leukocyte infiltration than the model group tissues (Table 2).

3.8. Effects of IA on protein expression in the paw tissues

In rat paw tissues, western blot analysis showed a significant increase in p-NF- κ B p65, p-I κ B α , p-JAK2, p-STAT3, p-JNK, p-EAR1/2 and p-38 in the model group compared with the control group. RIA (80, 40, 20 mg/kg) inhibited the phosphorylation of NF- κ B, JAK2/STAT3 and MAPK in rats. However, SIA (160, 80 mg/kg) down-regulated the levels of NF- κ B p65, p-I κ B α , p-JAK2, p-STAT3 and p-JNK in rat foot tissues, suggesting that IA had anti-inflammatory effects (Fig. 7).

3.9. HPLC-DAD-IT-TOF-MS profiles of *I. asprella* aqueous extracts

A total of 12 constituents were characterized by HPLC and LC-MS and confirmed by comparison with isolated pure compounds, including 10 phenolic acids are shown, neochlorogenic acid (1), 3,4,5-trimethoxyphenyl-1-*O*- β -D-apiofuranosyl-(1'→6')-glucopyranoside (2), cryptochlorogenic acid (3), chlorogenic acid (4), liriiodendrin (5), fraxiresinol 1-*O*- β -D-glucoside (6), (+)-syringaresinol β -D-glucoside (7), 3,4-dicaffeoylquinic acid (8), 3,5-dicaffeoylquinic acid (9), and 4,5-dicaffeoylquinic acid (10), and two triterpenoidal glucosides, asprellinoid B (11) and asprellinoid A (12). As no UV absorption was observed by the IA saponins, most of the peaks from triterpenoids and glucosides could be detected by LC-MS, as shown in Fig. 8B and Fig. 8D.

on a Waters Acquity UPLC BEH C18 column (100 mm $\times$ 3.0 mm, 1.7 μ m). Compounds 1: neochlorogenic acid; 2: 3,4,5-trimethoxyphenyl-1-*O*- β -D-apiofuranosyl-(1'→6')-glucopyranoside; 3: cryptochlorogenic acid; 4: chlorogenic acid; 5: liriiodendrin; 6: fraxiresinol 1-*O*- β -D-glucoside; 7: (+)-syringaresinol β -D-glucoside; 8: 3,4-dicaffeoylquinic acid; 9: 3,5-dicaffeoylquinic acid; 10: 4,5-dicaffeoylquinic acid; 11: asprellinoid B; 12: asprellinoid A.

Additionally, comparison of these chromatograms suggests a high similarity in the composition of RIA and SIA, except for the difference in chromatographic peaks between 30–37.5 min. This minor difference

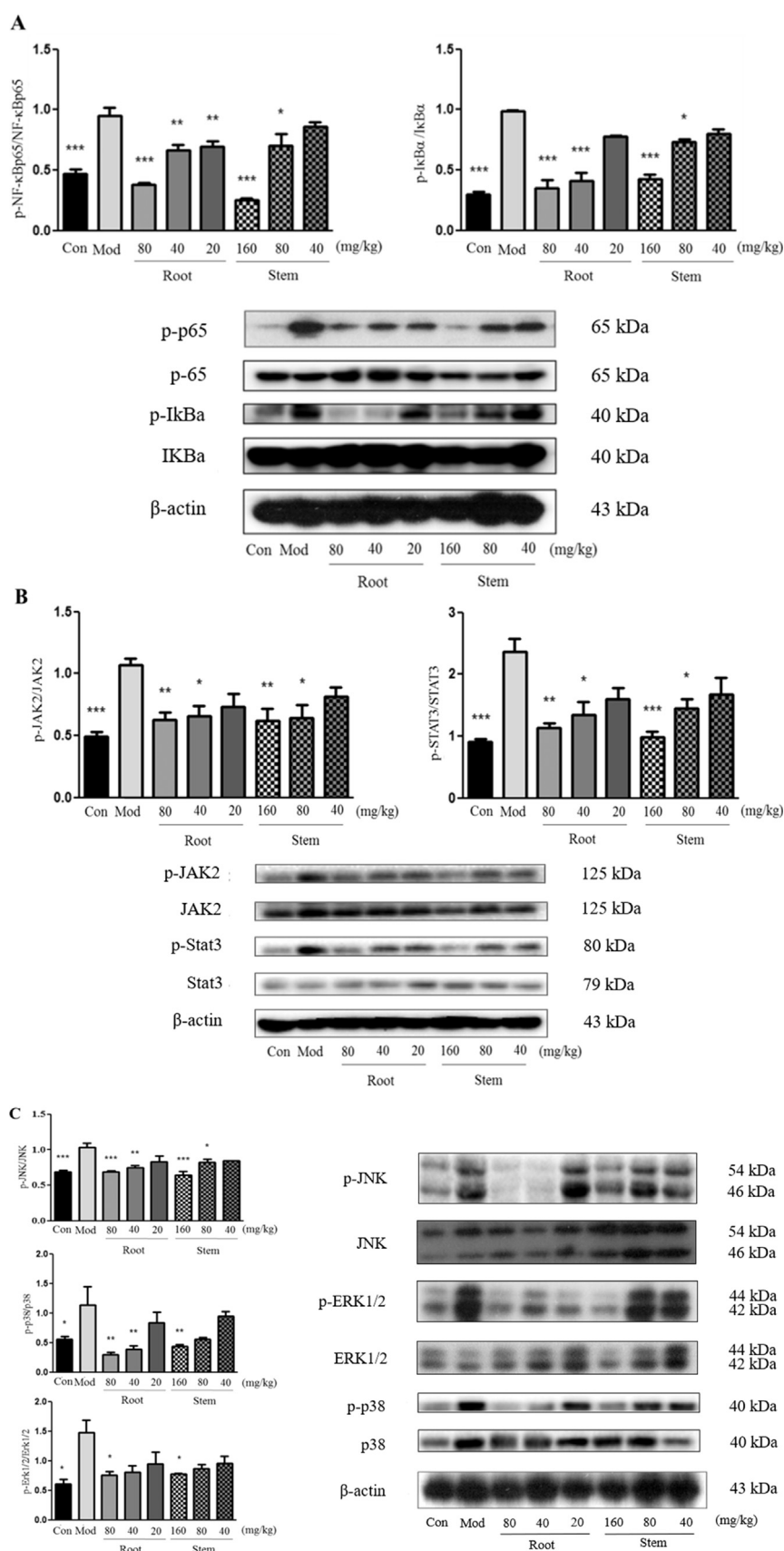


Fig. 7. Western blot analysis of NF-κB, JAK2/STAT3 and MAPK in the different treatment groups (A, B, C). The same site in foot tissues after carrageen treatment for 4 h in rats from different groups administered RIA (80, 40, 20 mg/kg) and SIA (160, 80, 40 mg/kg) was assessed by western blot analysis (n = 6). **P* < 0.05, ***P* < 0.01, and ****P* < 0.001 vs. the model group.

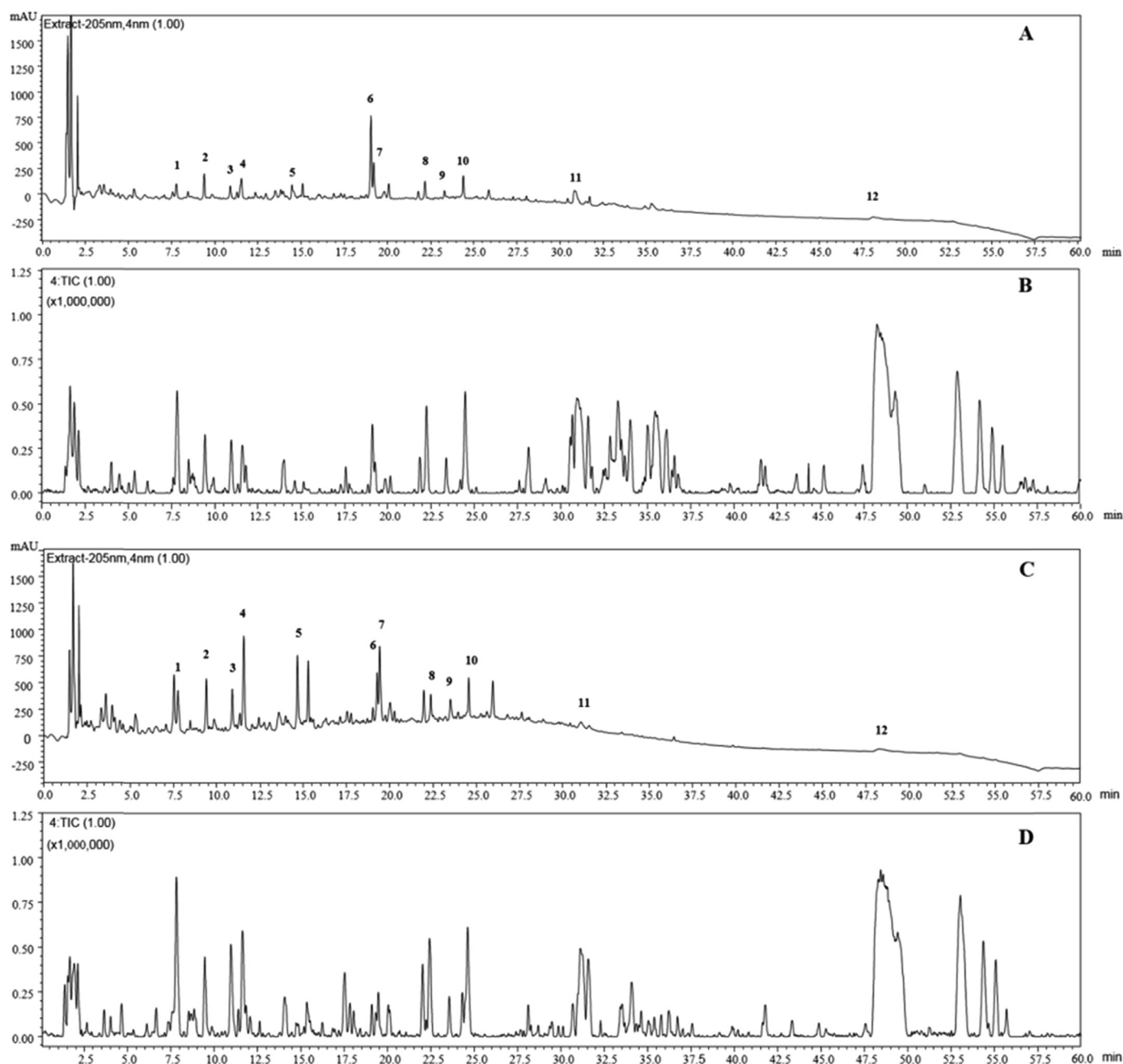


Fig. 8. HPLC-DAD (A) and LC-IT-TOFMS in negative mode (B) of RIA; HPLC-DAD (C) and LC-IT-TOFMS in negative mode (D) of SIA. The chromatographic separation was performed.

probably explains their anti-inflammatory activities at different dosages.

4. Discussion

The present study demonstrated the *in vivo* anti-inflammatory activity of RIA and SIA. The most effective doses of RIA (30 mg/kg) and SIA (60 and 30 mg/kg) were shown to have similar significant inhibitory effects on xylene-induced ear edema in mice, and RIA (80, 40 mg/kg) and SIA (160, 80 mg/kg) substantially decreased paw volume in rats. These dose-dependent activities of RIA and SIA were consistent with the reduced inflammatory cell infiltration and changes in the ear and paw tissues shown by HE staining and infiltration of neutrophil leukocytes in rat ear tissues shown by immunohistochemical analysis. Similarly, a decrease in the expression of key pro-

inflammatory factors, e.g., TNF- α , IL-6, and IL-1 β , in the serum of rats and mRNA levels of these genes in the paw tissues of rats from the RIA and SIA groups was observed, indicating the potential anti-inflammatory activity of these treatments. The present data showed that both RIA and SIA have anti-inflammatory effects, although these were present at different dosage ranges, which indicated that SIA may be used as an alternative of RIA in traditional anti-inflammatory Chinese herbal medicine in terms of the anti-inflammatory effect.

NF- κ B is an early nuclear transcription factor, and NF- κ B p65 is an important member of the NF- κ B transcription factor family, which plays a central role in the inflammatory response. This protein regulates the expression of downstream inflammatory factors, such as TNF, IL-6 and IL-1 β , and then participates in various inflammatory responses. The STAT3 signaling pathway also plays a key role in inflammatory responses. For example, IL-6 exerts its pro-inflammatory effects by

activating the STAT3 signaling pathway. TNF- α is primarily produced by mononuclear phagocytes and can induce immune responses by stimulating macrophages and neutrophils. This molecule can also induce the secretion of other inflammatory cytokines (Liao et al., 2013). Meanwhile, the MAPK pathway is similarly important in regulating inflammatory mediators, and the phosphorylation of p38 MAPK is an important event upstream of NF- κ B activation. Inhibitors of p38 MAPK effectively abolished phosphorylation of I κ B α and nuclear translocation of the RelA-p65 subunit of NF- κ B, thus preventing the transactivation mediated by NF- κ B. p38 MAPK can potentially affect the NF- κ B pathway through several mechanisms as shown as in previous studies (Baeza and Mun'oz-Ca'noves, 2004). For example, p38 is known to phosphorylate I κ B α , leading to its dissociation from the p65 subunit, thus facilitating the translocation of p65 to the nucleus (Calleros et al., 2006). Phosphorylated I κ B α was rapidly degraded by the proteasome (Finco et al., 1995) including JNK, ERK1/2 and p38, which is primarily involved in the regulation of inflammatory mediators. Activation of the MAPK pathway can promote the production of inflammatory cytokines and aggravate the inflammatory response (Wang et al., 2014). In this study, RIA and SIA reduced the phosphorylation levels of JAK2, STAT3, p65, I κ B α , p38, ERK1/2, and JNK in the rat paw tissues, suggesting the underlying mechanism of the RIA and SIA anti-inflammatory activities by inhibiting the production of key inflammatory mediators, including TNF- α , IL-6 and IL-1 β , through suppression of the NF- κ B, JAK2/STAT3 and MAPK signaling pathways.

Previous studies showed that IA contains phenolic acids, triterpenoids and glycosides as the major components (Dai et al., 2014; Du et al., 2017). Phenolic acids and triterpenoids from *Ilex* sp. were reported to have in vitro anti-inflammatory and antiviral activities and cytotoxicities; however, an in vivo evaluation has not been reported yet due to the difficulty in obtaining sufficient quantities of a single compound (Dai et al., 2014; Du et al., 2017). Chemical characterization by HPLC-DAD-IT-TOF-MS and comparison of the isolated compounds suggested that RIA and SIA contain a large number of constituents, but none are present at high levels, and that the anti-inflammatory effects are due to the contributions of phenolics or triterpenoids individually or synergistically (Zhao et al., 2013; Wang et al., 2014; Dai et al., 2014; Zhang et al., 2018). Additionally, RIA and SIA showed a high similarity in the composition of the constituents, except for some differences observed at the retention time of 30–37.5 min in the LC-MS chromatogram. These results suggest that SIA can be used as an alternative of RIA in terms of its anti-inflammatory effect.

Inflammation is a well-known pathophysiological factor of many diseases. The present study of IA, a folk herbal medicine in southern China, evaluated its anti-inflammatory effects and mechanisms and revealed its potential as a promising treatment for inflammatory diseases and a candidate for further development as an anti-inflammatory drug. This report also provided evidence supporting its traditional use for inflammation-related diseases, such as acute and chronic pharyngitis, cough, and sore throats, and suggested that SIA can be used as an alternative for RIA in terms of its anti-inflammatory effect. However, the bioactive ingredients that contribute to these effects, other pharmacological effects, and alternative uses of these treatments require further investigation.

5. Conclusion

The anti-inflammatory effects of *I. asprella* aqueous extracts were investigated by a xylene-induced ear edema test and carrageenan-induced rat paw edema test. The present results showed that both the root and stem of IA exerted significant anti-inflammatory activities in a dose-dependent manner (7.5–80 and 15–160 mg/kg, respectively). These effects were mediated by down-regulating the pro-inflammatory cytokines TNF- α , IL-6 and IL-1 β via inhibition of the NF- κ B, JAK2/STAT3 and MAPK signaling pathways. This is the first report of IA exerting anti-inflammatory effects in vivo, which provides a reference for

the treatment of inflammatory diseases, suggests IA should be further developed as an anti-inflammatory drug, supports its traditional use for inflammation-related diseases and indicates that SIA can be used as an alternative for RIA.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jep.2018.06.037.

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